INSTITUTE OF MATHEMATICAL SCIENCES, FACULTY OF SCIENCE, UNIVERSITI MALAYA
SEMESTER 2, SESSION 2025/2026
TUTORIAL 8

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1. Obtain the general solution for the following differential equations using method of undetermined coefficients:

   (a) $y'' - 4y' + 3y = e^{3x}$.

   (b) $y'' - y' - 2y = 4\sin x$.

   (c) $y'' - 3y' + 2y = 4xe^{x} + e^{2x}$.

   [Hint: $y_p = (Ax^2 + Bx)e^{x} + Cxe^{2x}$]

2. Given y_1 is a solution of the given differential equation. Using the method of reduction of order, find another solution y_2 such that y_1 and y_2 are linearly independent solutions of the differential equation.

   (a) $x^2y'' - xy' + y = 0, \quad y_1(x) = x$.

   (b) $(x+1)^2y'' - 2(x+1)y' + 2y = 0, \quad y_1(x) = x + 1$.

   (c) $xy'' + 2y' + xy = 0, \quad y_1(x) = \frac{\sin x}{x}$.

3. Obtain the general solution for the following differential equations using the method of variation parameters:

   (a) $y'' - 4y' + 4y = \frac{e^{2x}}{x}$.

   (b) $y'' + 2y' + 2y = e^{-x}\sec x$.